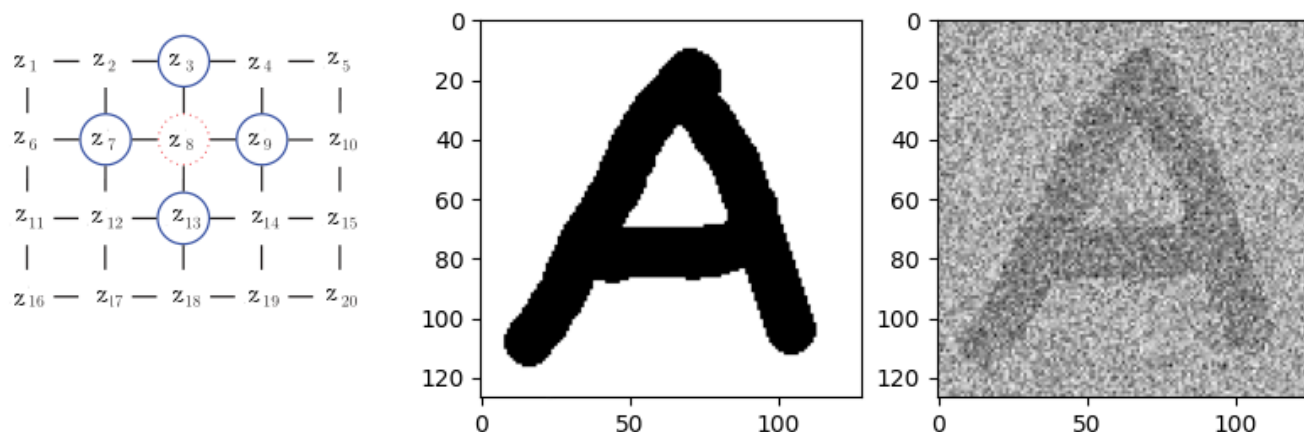


6. Markov Chain Monte Carlo (MCMC)

- 1 Stationarity
- 2 Metropolis-Hastings
- 3 Gibbs Sampling
- 4 Conclusion
- 5 Some Practice Problems

Motivation



$$p(\mathbf{z} \mid \mathbf{y}) = \frac{p(\mathbf{y} \mid \mathbf{z})p(\mathbf{z})}{p(\mathbf{y})}$$

- For good performance, need a “reasonable” prior $p(\mathbf{z})$ that models relationship between neighboring pixels.
- But this makes $p(\mathbf{z} \mid \mathbf{y})$ a **high dimensional** distribution, not possible to have analytical form.
- We want to sample from $p(\mathbf{z} \mid \mathbf{y})$!

Introduction

- We have studied various sampling methods: from **standard distributions**, **rejection sampling**, and **importance sampling**.
- These can be used for **low dimensional pdfs** (e.g., scalar valued random variables) and as **subroutines** in other sampling methods.
- Do not work well in high dimensional spaces.
- **Markov Chain Monte Carlo (MCMC)** is the method for sampling from high-dimensional distributions.
- Regarded as one of the **top 10 most important algorithms of the 20th century**.
- History:
 - ▶ Metropolis et al., 1953 (physics and chemistry)
 - ▶ Hastings, 1970 (statistics, still relatively unknown)
 - ▶ Geman and Geman, 1984 (Gibbs sampling)
 - ▶ Gelfand and Smith, 1990 (popularize MCMC)
- Relies on the concept of “**stationary distribution**” of a Markov chain.
 - ▶ **Want to sample from a target pdf $\pi(\mathbf{x})$.**
 - ▶ **Design a MC $Z_0, Z_1, \dots$ such that distribution of Z_n approaches $\pi(\cdot)$ as $n \rightarrow \infty$.**

Outline

- 1 Stationarity
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Stationary Distribution

- Consider a homogeneous Markov chain with transition probability $p(x_t = y \mid x_{t-1} = x) = \mathbf{T}(x, y)$.
- π is a **stationary distribution** if

$$\sum_x \pi(x) \mathbf{T}(x, y) = \pi(y)$$

for all states y .

- Also called invariant distribution - distribution does not change as MC proceeds.
- If there are M states,

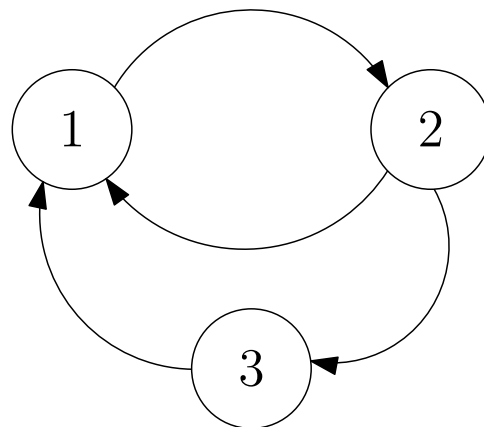
$$\mathbf{T} = \begin{bmatrix} \mathbf{T}(1, 1) \cdots \mathbf{T}(1, M) \\ \vdots \\ \mathbf{T}(M, 1) \cdots \mathbf{T}(M, M) \end{bmatrix}$$

- Stationary distribution satisfies $\pi \mathbf{T} = \pi$.

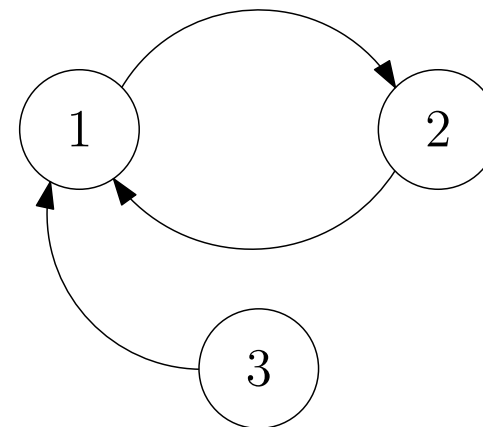
Conditions for Stationarity (Supplementary)

- Conditions for existence of stationary distribution:

- ▶ Markov chain is **irreducible**: there is a **positive probability** of going from one state to another state in a finite number of steps. Ensures that **every state is visited infinitely often**.



irreducible

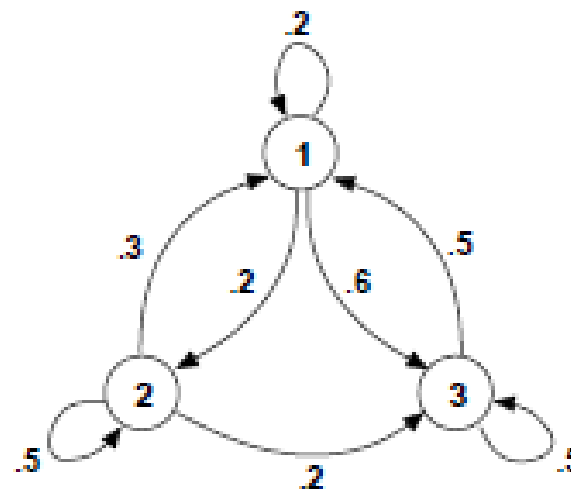


reducible

- ▶ Markov chain is **positive recurrent**: starting in any state i , the **expected time to return** to state i is **finite**. Equivalently, the proportion of time spent in every state is positive as length of chain $n \rightarrow \infty$. Ensures that no one state is “left out”.

- Example: 3 state chain with transition matrix

$$\mathbf{T} = \begin{bmatrix} 0.2 & 0.2 & 0.6 \\ 0.3 & 0.5 & 0.2 \\ 0.5 & 0.0 & 0.5 \end{bmatrix}$$



Solve $\pi \mathbf{T} = \pi \quad \Rightarrow \quad \pi = \left[\frac{25}{69}, \frac{10}{69}, \frac{34}{69} \right]$.

- Fact: a **finite state** irreducible Markov chain is always positive recurrent, therefore it has a stationary distribution.

Asymptotic Steady State

- As k increases, will the Markov chain reach a “steady state” regardless of initial distribution?
- π_0 - initial distribution, $\pi_1 = \pi_0 \mathbf{T}$, $\pi_2 = \pi_1 \mathbf{T}$, $\dots$, $\pi_k = \pi_{k-1} \mathbf{T} = \pi_0 \mathbf{T}^k$
- Does $\pi = \lim_{k \rightarrow \infty} \pi_k = \lim_{k \rightarrow \infty} \pi_0 \mathbf{T}^k$ exist?
- If it does, π must be a stationary distribution.

Asymptotic Steady State (Supplementary)

- Conditions for $\pi = \lim_{k \rightarrow \infty} \pi_0 \mathbf{T}^k$:
 - ▶ Markov chain is **irreducible** and **positive recurrent**: stationary distribution π exists.
 - ▶ Markov chain is **aperiodic**: suppose we have two states 1 and 2, with $\mathbf{T}(1, 2) = \mathbf{T}(2, 1) = 1$, i.e., when in state 1, always transition to state 2, and vice versa. We see that this chain oscillates between states 1 and 2. Its period is 2. This MC does not converge.

$$\mathbf{T}^k = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{for } k \text{ odd}, \quad \mathbf{T}^k = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{for } k \text{ even}$$

When a Markov chain is aperiodic, it means that the period is 1.

- Such a MC is said to be **ergodic**.
- We will always assume above conditions hold.
- After a sufficiently long **burn-in** period m , the distribution at each time $k > m$ becomes approximately the stationary distribution.

Reversible MC

- *Sufficient* but not necessary condition for π to be stationary is

$$\pi(x)\mathbf{T}(x, y) = \pi(y)\mathbf{T}(y, x)$$

for all x, y . This Markov chain is said to be *reversible*.

$$\sum_x \pi(x)\mathbf{T}(x, y) = \sum_x \pi(y)\mathbf{T}(y, x) = \pi(y) \sum_x \mathbf{T}(y, x) = \pi(y)$$

- Suppose we want to sample from a distribution $\pi(x)$. We design a transition probability $\mathbf{T}(x, y)$ such that
 - ▶ all the ergodicity conditions are satisfied
 - ▶ (usually make the Markov chain reversible and aperiodic)
 - ▶ $\pi(x)$ is its stationary distribution.
- We can generate a sample path for the Markov chain $z_0, z_1, \dots, z_n$, starting from **any** z_0 . If n is large, we have $p(z_n) \approx \pi(z_n)$, so we will have obtained a **sample from π !** **This is the basis of MCMC.**

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- 1 Stationarity
- 2 Metropolis-Hastings
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Metropolis-Hastings Algorithm

- Want to sample from $\pi(\mathbf{x})$, $\mathbf{x} \in \mathcal{X}$ - hard. [E.g., $\mathcal{X} = \mathbb{R}^{1000}$.]
- Assume we can compute $\tilde{\pi}(\mathbf{x})$, which is the un-normalized version of $\pi(\mathbf{x})$.
- Choose a **transition probability** $q(\mathbf{x}, \mathbf{y}) = q(\mathbf{y} \mid \mathbf{x})$, *irreducible and aperiodic* on $\mathcal{X}$, and easy to sample from. This is called the **proposal distribution**.

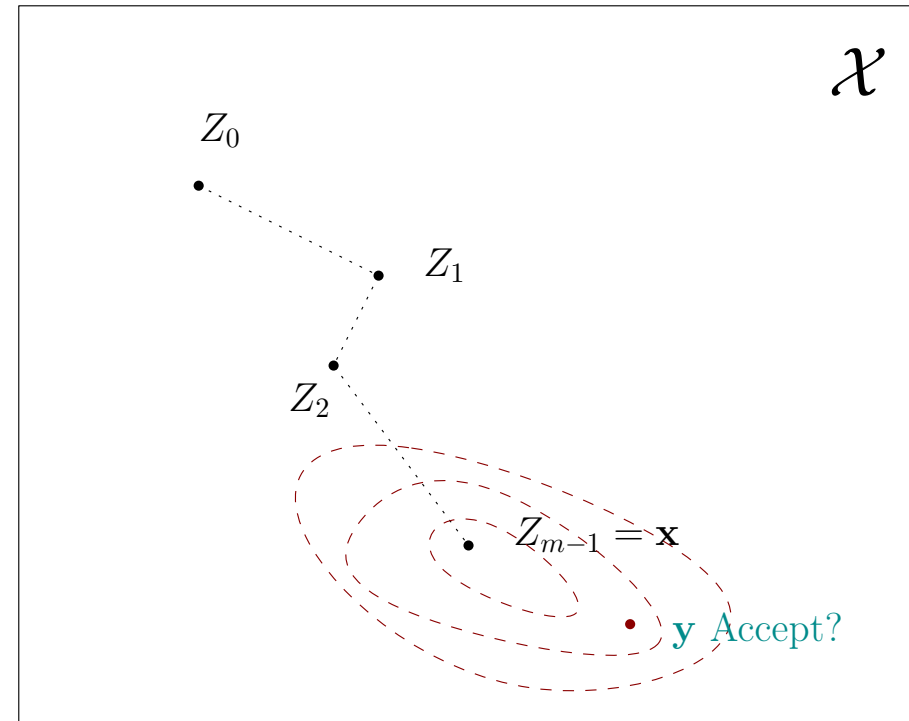
Let Z_m , $m = 0, 1, \dots$ be the realizations of the following Markov chain.

At step m ,

1. Let $\mathbf{x} = Z_{m-1}$.
2. Sample $\mathbf{y}$ from $q(\mathbf{x}, \mathbf{y})$.
3. Accept $\mathbf{y}$ with probability

$$A(\mathbf{x}, \mathbf{y}) = \min \left(1, \frac{\tilde{\pi}(\mathbf{y})q(\mathbf{y}, \mathbf{x})}{\tilde{\pi}(\mathbf{x})q(\mathbf{x}, \mathbf{y})} \right)$$

4. If $\mathbf{y}$ is accepted, set $Z_m = \mathbf{y}$.
Otherwise, set $Z_m = \mathbf{x} = Z_{m-1}$.



Metropolis-Hastings Algorithm

- $Z_0, Z_1, \dots$ is a Markov chain as Z_m depends only on the value of Z_{m-1} .
- Similar to rejection and importance sampling, no need to know normalizing constant for $\tilde{\pi}(\mathbf{x})$.
- Suitable for high dimensional distributions as sampling is from an “easy” proposal $q(\mathbf{x}, \cdot)$.

Proof of MH (Supplementary)

Lemma

The constructed MH Markov chain is reversible and $\pi(\mathbf{x})$ is its stationary distribution.

Proof.

Let $T(\mathbf{x}, \mathbf{y})$ be the transition probability of the MC. We show that the MC is reversible w.r.t. $\pi(\mathbf{x})$. That means $\pi(\mathbf{x})$ is the stationary distribution. For $\mathbf{x} \neq \mathbf{y}$,

$$\begin{aligned}\pi(\mathbf{x})T(\mathbf{x}, \mathbf{y}) &= \pi(\mathbf{x})q(\mathbf{x}, \mathbf{y})A(\mathbf{x}, \mathbf{y}) = \pi(\mathbf{x})q(\mathbf{x}, \mathbf{y}) \min\left(1, \frac{\pi(\mathbf{y})q(\mathbf{y}, \mathbf{x})}{\pi(\mathbf{x})q(\mathbf{x}, \mathbf{y})}\right) \\ &= \min(\pi(\mathbf{x})q(\mathbf{x}, \mathbf{y}), \pi(\mathbf{y})q(\mathbf{y}, \mathbf{x})) = \pi(\mathbf{y})q(\mathbf{y}, \mathbf{x}) \min\left(\frac{\pi(\mathbf{x})q(\mathbf{x}, \mathbf{y})}{\pi(\mathbf{y})q(\mathbf{y}, \mathbf{x})}, 1\right) \\ &= \pi(\mathbf{y})T(\mathbf{y}, \mathbf{x})\end{aligned}$$

□

- If the MH Markov chain is irreducible, (positive recurrent,) aperiodic $\Rightarrow$ distribution of Z_n approaches $\pi(\cdot)$ as $n \rightarrow \infty$.
- E.g., if state space $\mathcal{X}$ is finite, choose $q(\mathbf{x}, \mathbf{y}) > 0, \forall \mathbf{x}, \mathbf{y} \in \mathcal{X}$.

Proposal Distributions

- MH Markov chain $Z_0, Z_1, \dots$ is designed to converge to the stationary distribution $\pi(\cdot)$.
- I.e., the samples Z_m for large m are approximate samples from $\pi(\cdot)$.
- Run the MC for a large number of steps and discard these. This is the burn-in period. Typically, the first 1000 to 5000 samples are discarded.

Proposal Distributions

- Choice of proposal distribution can affect performance of the algorithm. Some typical choices are:
 - ▶ $q(\mathbf{x}, \mathbf{y}) = q(\mathbf{y} - \mathbf{x})$ - random walk MH
 - $\mathbf{y} - \mathbf{x} \sim \mathcal{N}(\cdot | \mathbf{0}, \Sigma)$ - Gaussian centered at $\mathbf{x}$.
 - $\mathbf{y} - \mathbf{x} \sim \text{Unif}[-\delta, \delta]^d$ - Uniform distribution centered at $\mathbf{x}$

Remark 1: In these examples, $q(\mathbf{x}, \mathbf{y}) = q(\mathbf{y}, \mathbf{x}) \Rightarrow$

$$A(\mathbf{x}, \mathbf{y}) = \min \left(1, \frac{\tilde{\pi}(\mathbf{y})q(\mathbf{y}, \mathbf{x})}{\tilde{\pi}(\mathbf{x})q(\mathbf{x}, \mathbf{y})} \right) = \min \left(1, \frac{\tilde{\pi}(\mathbf{y})}{\tilde{\pi}(\mathbf{x})} \right)$$

This is known as the Metropolis Algorithm.

Remark 2: Random walk in the state space. Need to choose the variance appropriately. If variance is small, progress through the state space is slow. If variance is high, we have high rejection rates.

Proposal Distributions

- ► $q(\mathbf{x}, \mathbf{y}) = q(\mathbf{y})$ - **independence chain** MH, choose the next state independently of the current state $\mathbf{x}$.
 - Depends on how closely proposal distribution $q(\mathbf{y})$ approximates $\pi(\mathbf{y})$.
 - In general, behaves well only if $q(\mathbf{y})$ is more heavy-tailed than $\pi(\mathbf{y})$.
- ► **Exploit properties** of $\pi(\mathbf{x})$. Suppose $\pi(\mathbf{x}) \propto \psi(\mathbf{x})h(\mathbf{x})$, where $h(\mathbf{x})$ is a density that can be sampled from, and $\psi(\mathbf{x})$ is a bounded function. Take $q(\mathbf{x}, \mathbf{y}) = h(\mathbf{y})$. Then,

$$\begin{aligned} A(\mathbf{x}, \mathbf{y}) &= \min \left(1, \frac{\tilde{\pi}(\mathbf{y})q(\mathbf{y}, \mathbf{x})}{\tilde{\pi}(\mathbf{x})q(\mathbf{x}, \mathbf{y})} \right) \\ &= \min \left(1, \frac{\psi(\mathbf{y})h(\mathbf{y})h(\mathbf{x})}{\psi(\mathbf{x})h(\mathbf{x})h(\mathbf{y})} \right) \\ &= \min \left(1, \frac{\psi(\mathbf{y})}{\psi(\mathbf{x})} \right). \end{aligned}$$

Example

$\pi(x) \propto \exp(-x^2)x^2(5 + \sin(2x))$. Use a Gaussian transition proposal distribution (random walk MH):

$$q(x, y) = \mathcal{N}(y \mid x, \sigma^2) = \mathcal{N}(x \mid y, \sigma^2) = q(y, x)$$

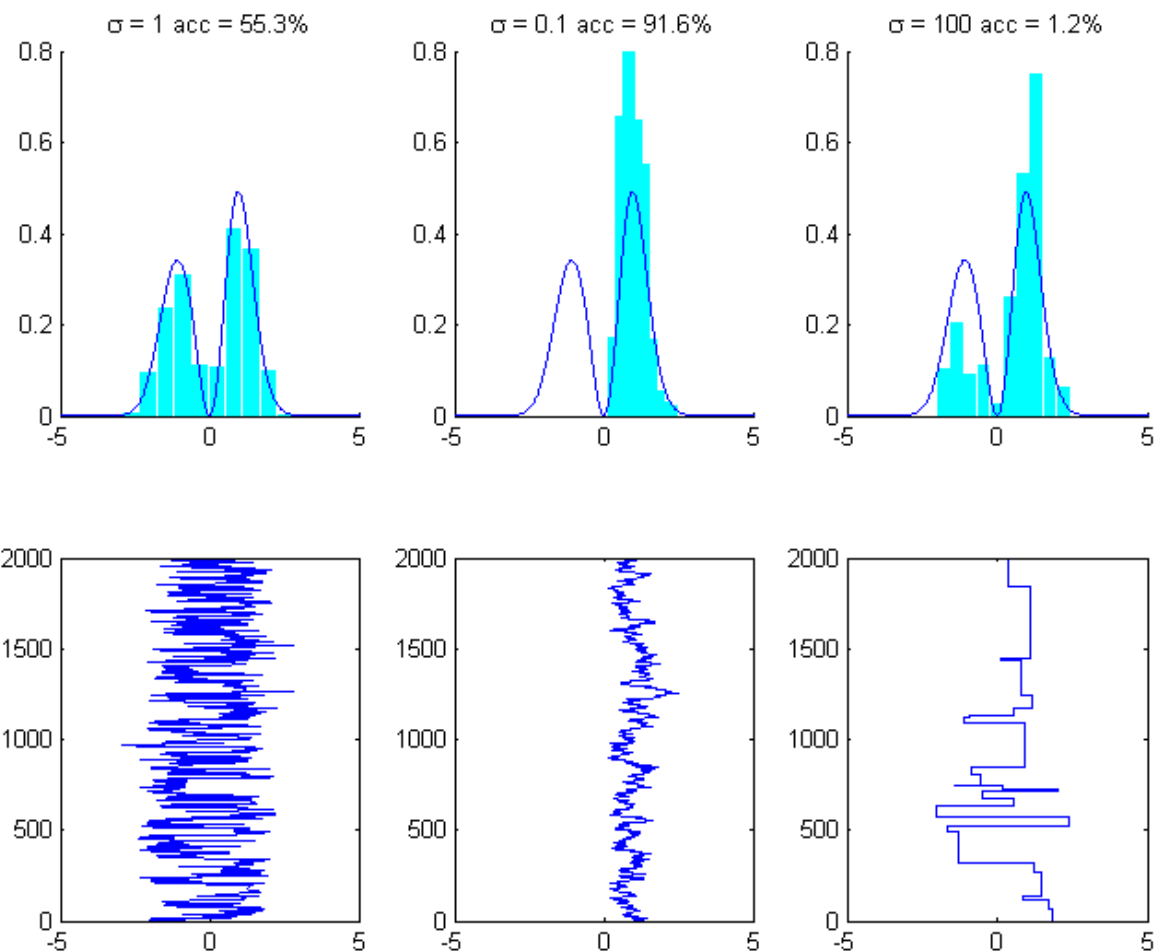
so

$$A(x, y) = \min \left(1, \frac{\tilde{\pi}(y)}{\tilde{\pi}(x)} \right)$$

At each step,

- Sample y from $q(Z_m, y)$.
- Generate uniform random variable $U \sim \text{Unif}[0, 1]$. Set $Z_{m+1} = y$ if $U \leq A(Z_m, y)$. Otherwise, set $Z_{m+1} = Z_m$.

Example

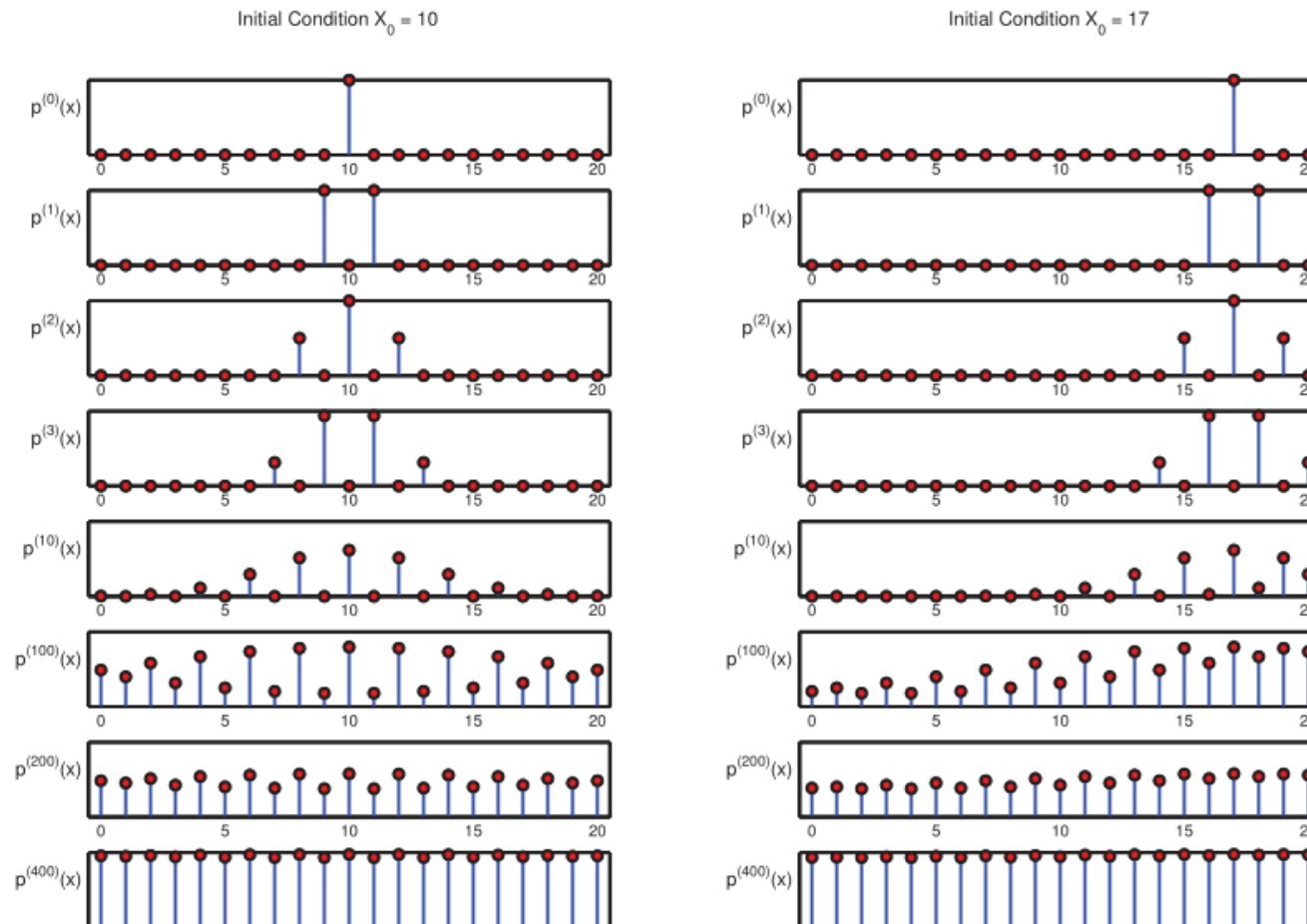


Proposal Variance

- Important to use appropriate σ .
- σ too small - slow convergence, takes a long time to explore the whole space, acceptance rate too high.
- σ too large - big steps, but very low acceptance rate, gets stuck for a long time.
- Rules of thumb:
 - ▶ random walk MH: acceptance rate of 0.25 to 0.5
 - ▶ independence chain MH: acceptance rate close to 1

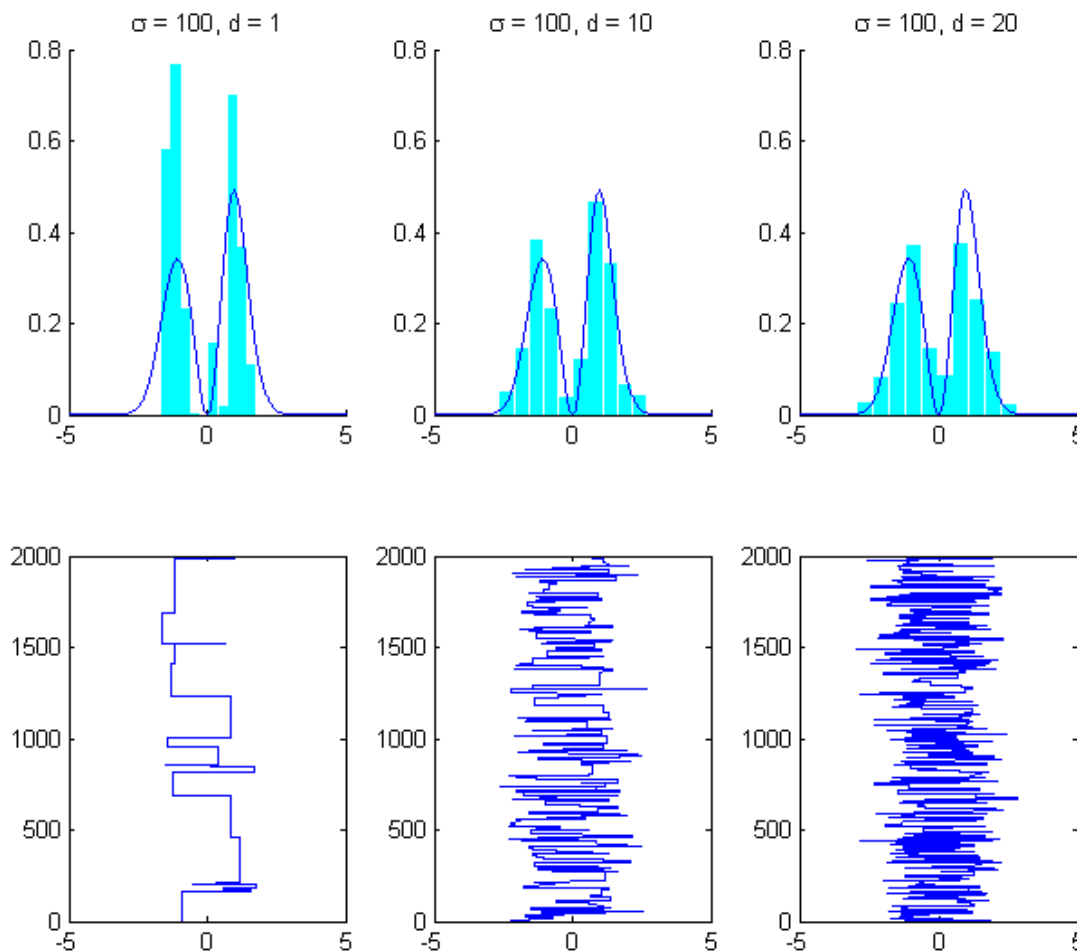
Burn-In

- **Burn-in** phase: throw away samples from initial period before chain is close to the target stationary distribution.
- But **difficult to diagnose** when the chain has burned in. E.g., $\text{Unif}(\{0, 1, \dots, 20\})$. Takes over 400 steps until chain has “forgotten” its starting point.



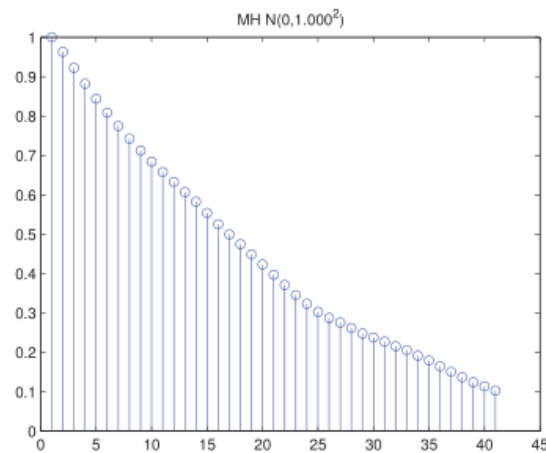
Thinning

- **Thinning** - break dependence between draws in the MC by taking every d th sample.
- Useful when σ is chosen too large - stuck at same location for long periods of time. Sub-sample every d samples from original MC.

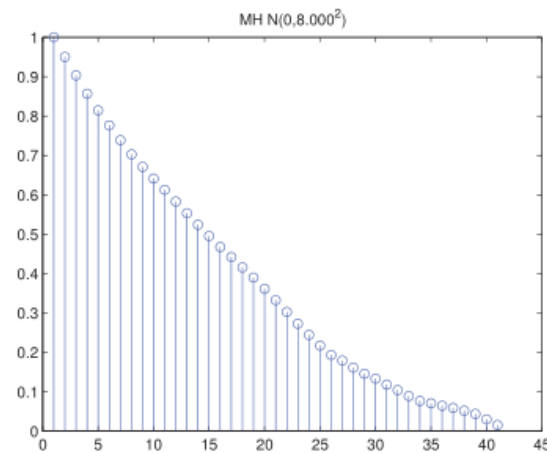


Python

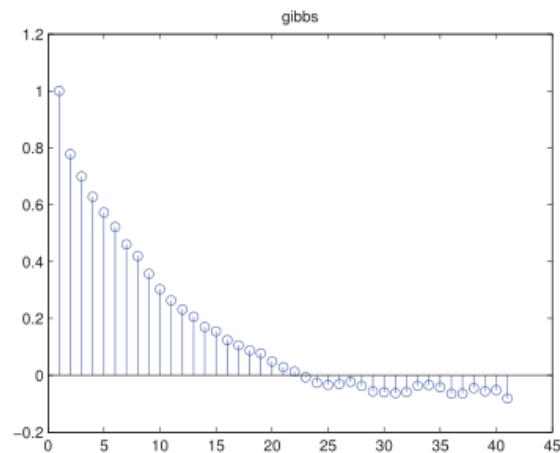
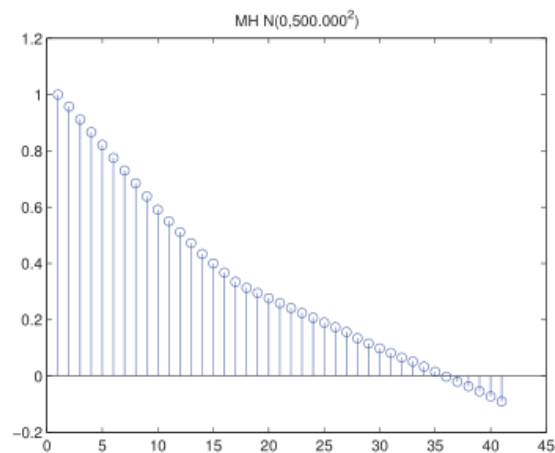
- [M12]: 06_mcmc_gmm_demo.ipynb
- Use thinning: subsample every d samples.
- Larger $d \Rightarrow$ lower correlation between samples.
- Tradeoff: more samples wasted.



(a)



(b)



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Gibbs Sampling

- Special case of Metropolis-Hastings.
- We want to sample from a multivariate distribution $p(z_1, \dots, z_d)$ - hard to do so directly if dimension is large.
- For each i , let $\mathbf{z}_{-i} = \{z_1, \dots, z_{i-1}, z_{i+1}, \dots, z_d\}$, i.e., z_i removed.
- If we can compute the **full conditionals** $p(z_i | \mathbf{z}_{-i})$, then we can use the Gibbs sampler:
 - ▶ Initialize $(z_1^{(0)}, \dots, z_d^{(0)})$.
 - ▶ For each k :
 - sample $z_1^{(k)}$ from $p(\cdot | z_2^{(k-1)}, \dots, z_d^{(k-1)})$
 - sample $z_2^{(k)}$ from $p(\cdot | z_1^{(k)}, z_3^{(k-1)}, \dots, z_d^{(k-1)})$
 - $\vdots$
 - sample $z_j^{(k)}$ from $p(\cdot | z_1^{(k)}, \dots, z_{j-1}^{(k)}, z_{j+1}^{(k-1)}, \dots, z_d^{(k-1)})$
 - $\vdots$
 - sample $z_d^{(k)}$ from $p(\cdot | z_1^{(k)}, \dots, z_{d-1}^{(k)})$
 - ▶ Burn-in period samples discarded.

Gibbs Sampling: Why it works (Supplementary)

- In iteration k and the step of the Gibbs sampler where we sample $z_j^{(k)}$, let

$$\mathbf{x}_j^{(k)} = (z_1^{(k)}, \dots, z_{j-1}^{(k)}, z_j^{(k-1)}, z_{j+1}^{(k-1)}, \dots, z_d^{(k-1)}),$$

$$\mathbf{x}_{-j}^{(k)} = (z_1^{(k)}, \dots, z_{j-1}^{(k)}, z_{j+1}^{(k-1)}, \dots, z_d^{(k-1)}),$$

$$\mathbf{y}_j^{(k)} = (z_1^{(k)}, \dots, z_{j-1}^{(k)}, z_j^{(k)}, z_{j+1}^{(k-1)}, \dots, z_d^{(k-1)}).$$

- Suppose $\mathbf{x}_j^{(k)} \sim p(z_1, \dots, z_d)$, i.e., $p_{\mathbf{x}_{-j}^{(k)}}(\mathbf{x}_{-j}^{(k)}) = p(\mathbf{z}_{-j} = \mathbf{x}_{-j}^{(k)})$.
- We sample $z_j^{(k)}$ from $p(z_j \mid \mathbf{x}_{-j}^{(k)})$ and $\mathbf{x}_{-j}^{(k)}$ is unchanged,

$$\begin{aligned} p_{\mathbf{y}_j^{(k)}}(\mathbf{y}_j^{(k)}) &= p(z_j = z_j^{(k)} \mid \mathbf{x}_{-j}^{(k)}) p_{\mathbf{x}_{-j}^{(k)}}(\mathbf{x}_{-j}^{(k)}) \\ &= p(z_j = z_j^{(k)} \mid \mathbf{x}_{-j}^{(k)}) p(\mathbf{z}_{-j} = \mathbf{x}_{-j}^{(k)}) \\ &= p(\mathbf{z} = \mathbf{y}_j^{(k)}). \end{aligned}$$

- Therefore, $\mathbf{y}_j^{(k)}$ has joint pdf $p(z_1, \dots, z_d)$.

Gibbs Sampling: Why it works (Supplementary)

$$\begin{aligned}\mathbf{x}_j^{(k)} &= (z_1^{(k)}, \dots, z_{j-1}^{(k)}, z_j^{(k-1)}, z_{j+1}^{(k-1)}, \dots, z_d^{(k-1)}), \\ \mathbf{x}_{-j}^{(k)} &= (z_1^{(k)}, \dots, z_{j-1}^{(k)}, z_{j+1}^{(k-1)}, \dots, z_d^{(k-1)}), \\ \mathbf{y}_j^{(k)} &= (z_1^{(k)}, \dots, z_{j-1}^{(k)}, z_j^{(k)}, z_{j+1}^{(k-1)}, \dots, z_d^{(k-1)}).\end{aligned}$$

- Equivalent to MH step with transition from $\mathbf{x}_j^{(k)}$ to $\mathbf{y}_j^{(k)}$ given by transition probability $q_j(\mathbf{x}_j^{(k)}, \mathbf{y}_j^{(k)}) = p(z_j^{(k)} | \mathbf{x}_{-j}^{(k)})$ and acceptance probability

$$\begin{aligned}A(\mathbf{x}_j^{(k)}, \mathbf{y}_j^{(k)}) &= \frac{p(\mathbf{y}_j^{(k)})q_j(\mathbf{y}_j^{(k)}, \mathbf{x}_j^{(k)})}{p(\mathbf{x}_j^{(k)})q_j(\mathbf{x}_j^{(k)}, \mathbf{y}_j^{(k)})} = \frac{p(\mathbf{y}_j^{(k)})p(z_j^{(k-1)} | \mathbf{x}_{-j}^{(k)})}{p(\mathbf{x}_j^{(k)})p(z_j^{(k)} | \mathbf{x}_{-j}^{(k)})} \\ &= \frac{p(z_j^{(k)} | \mathbf{x}_{-j}^{(k)})p(\mathbf{x}_{-j}^{(k)})p(z_j^{(k-1)} | \mathbf{x}_{-j}^{(k)})}{p(z_j^{(k-1)} | \mathbf{x}_{-j}^{(k)})p(\mathbf{x}_{-j}^{(k)})p(z_j^{(k)} | \mathbf{x}_{-j}^{(k)})} = 1.\end{aligned}$$

Gibbs Sampling

- Procedures to get approximately i.i.d. samples:
 - ▶ Generate r independent Gibbs sequences of length m , and take the final value from each sequence.
 - ▶ Generate one long Gibbs sequence, discard burn-in period, and take every d th sample.
- How to get conditional distributions?

$$p(z_1 \mid z_2, \dots, z_d) = \frac{p(z_1, \dots, z_d)}{p(z_2, \dots, z_d)}.$$

- ▶ Start with joint distribution $p(z_1, \dots, z_d)$.
- ▶ Ignore everything that does not depend on z_1 .
- ▶ Use knowledge of well-known distributions to find the distribution $p(z_1 \mid z_2, \dots, z_d)$.

Example

Gibbs sampling of the following pdf:

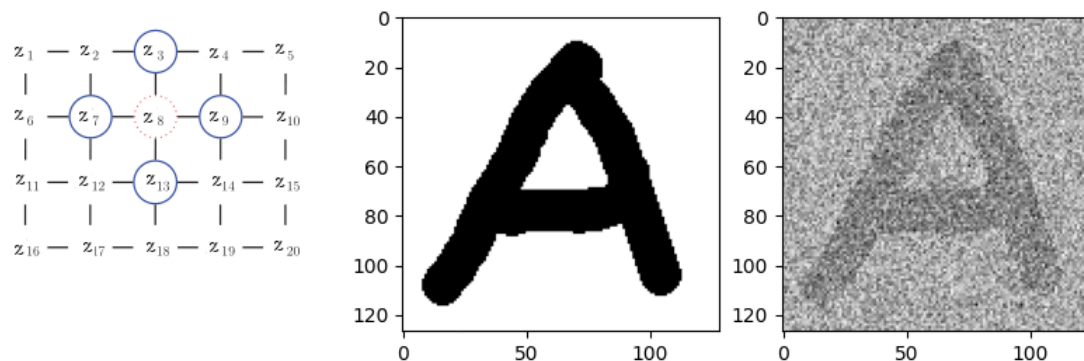
$$p(x, y, n) \propto \binom{n}{x} y^{x+a-1} (1-y)^{n-x+b-1} e^{-\lambda} \frac{\lambda^n}{n!}$$

where a, b, λ are positive constants.

- $p(x | y, n) \propto \binom{n}{x} y^x (1-y)^{n-x}$ is $\text{Bin}(x | n, y)$
- $p(y | x, n) \propto y^{x+a-1} (1-y)^{n-x+b-1}$ is $\text{Beta}(y | x+a, n-x+b)$
- $p(n | x, y) \propto \frac{((1-y)\lambda)^{n-x}}{(n-x)!}$ is $\text{Pois}(n | (1-y)\lambda) + x$
- Even if we just want to sample x ,
 - ▶ use the Gibbs sampler to sample from the joint distribution $p(x, y, n)$
 - ▶ discard samples of y and n
- This is because it is **hard to find** $p(x)$ directly!

A Real Application: Denoising using the Ising Model

- Image denoising: for a white/black image, each pixel $z_j \in \{-1, 1\}$ is a hidden value of the clean image.



- Given a noisy image $\mathbf{y}$, want to recover $\mathbf{z}$. **Bayesian inference: compute the posterior $p(\mathbf{z} \mid \mathbf{y})$.**
- Likelihood: $p(\mathbf{y} \mid \mathbf{z}) = \prod_j p(y_j \mid z_j) = \prod_j \mathcal{N}(y_j \mid z_j, \sigma^2)$, i.e., each pixel is independently corrupted with Gaussian noise.
- Ising prior distribution:

$$p(z_j \mid \mathbf{z}_{-j}) \propto \prod_{s \in N_j} \psi(z_s, z_j),$$

where N_j is the neighborhood of pixel z_j , and $\psi(u, v) = \exp(Juv)$ with $J > 0$ as the “coupling strength”.

A Real Application: Denoising using the Ising Model

Let's compute the Ising prior distribution (we will need it later):

$$\begin{aligned} p(z_j \mid \mathbf{z}_{-j}) &= \frac{\prod_{s \in N_j} \psi(z_s, z_j)}{\prod_{s \in N_j} \psi(z_s, z_j) + \prod_{s \in N_j} \psi(z_s, -z_j)} \\ &= \frac{\exp(J \sum_{s \in N_j} z_s z_j)}{\exp(J \sum_{s \in N_j} z_s z_j) + \exp(-J \sum_{s \in N_j} z_s z_j)} \\ &= \frac{1}{1 + \exp(-2J \sum_{s \in N_j} z_s z_j)} \\ &= \text{sigmoid}(2J\eta_j), \end{aligned}$$

where $\text{sigmoid}(u) = \frac{1}{1 + \exp(-u)}$ and $\eta_j = \sum_{s \in N_j} z_s z_j$.

A Real Application: Denoising using the Ising Model

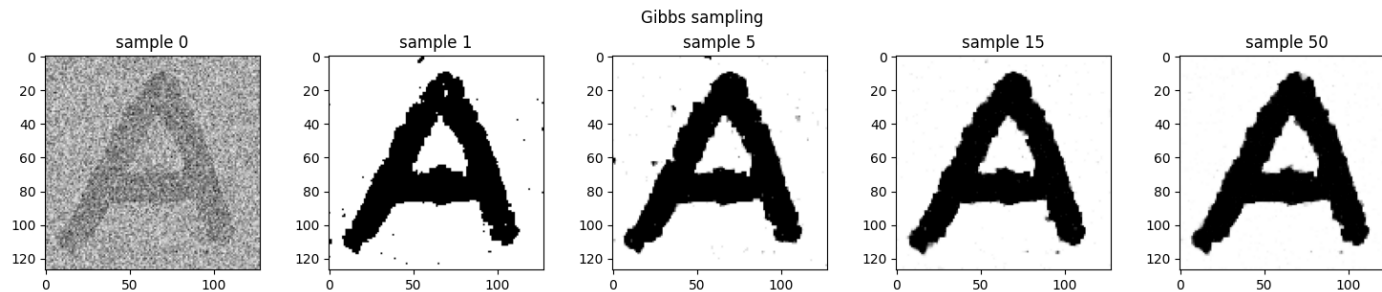
- We want to sample from $p(\mathbf{z} \mid \mathbf{y})$.
- Gibbs sampling: need the conditional distributions

$$\begin{aligned}
 p(z_j \mid \mathbf{z}_{-j}, \mathbf{y}) &= p(z_j \mid \mathbf{z}_{-j}, y_j) \quad \text{since } y_j \text{ depends only on } z_j \\
 &= \frac{p(z_j \mid \mathbf{z}_{-j})p(y_j \mid z_j)}{p(y_j \mid \mathbf{z}_{-j})} \\
 &= \frac{\text{sigmoid}(2J\eta_j)p(y_j \mid z_j)}{\text{sigmoid}(2J\eta_j)p(y_j \mid z_j) + \text{sigmoid}(-2J\eta_j)p(y_j \mid -z_j)} \\
 &= \text{sigmoid} \left(2J\eta_j - \log \frac{\mathcal{N}(y_j \mid -z_j, \sigma^2)}{\mathcal{N}(y_j \mid z_j, \sigma^2)} \right)
 \end{aligned}$$

- Reminder: z_j is binary, so it is easy to sample using the above conditional distribution.

A Real Application: Denoising using the Ising Model

- [M12]: 06_ising_image_denoise_demo.ipynb: interested to find the mean of the posterior distribution $p(\mathbf{z} \mid \mathbf{y})$.
- Gibbs sampling: sample from $p(\mathbf{z} \mid \mathbf{y})$, and take the mean of the samples as an estimate of the posterior mean.



- Bayesian neural networks: MCMC Training of Bayesian Neural Networks

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Summary

- Stationarity of Markov chains: irreducible and positive recurrent
- Asymptotic convergence: irreducible, positive recurrent and aperiodic
- Metropolis-Hastings algorithm
- Gibbs sampling
- [06_sampling_summary.ipynb](#)

“... from now on we can compare our data with the model we actually want to use rather than with a model which has some mathematical convenient form. This is surely a revolution.”

Peter Clifford (1993)

Outline

- 1 Stationarity
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Some Practice Problems

1. Show that in MH sampling, if we were able to sample from $q(x, y) = \pi(y)$, then MH sampling reduces to ordinary sampling.

Solution: At each step, we sample directly from π with acceptance probability

$$A(x, y) = \min\left(1, \frac{\pi(y)q(y, x)}{\pi(x)q(x, y)}\right) = \min\left(1, \frac{\pi(y)\pi(x)}{\pi(x)\pi(y)}\right) = 1.$$

2. Consider $p(z_1, z_2, z_3) \propto z_1^{z_2+a-2}(1-z_1)^{b-1} \frac{1}{z_2!} \mathbf{1}_{\{z_3 \in [0, z_1]\}}$. Derive all the conditional probabilities used in the Gibbs sampling of (z_1, z_2, z_3) .

Solution:

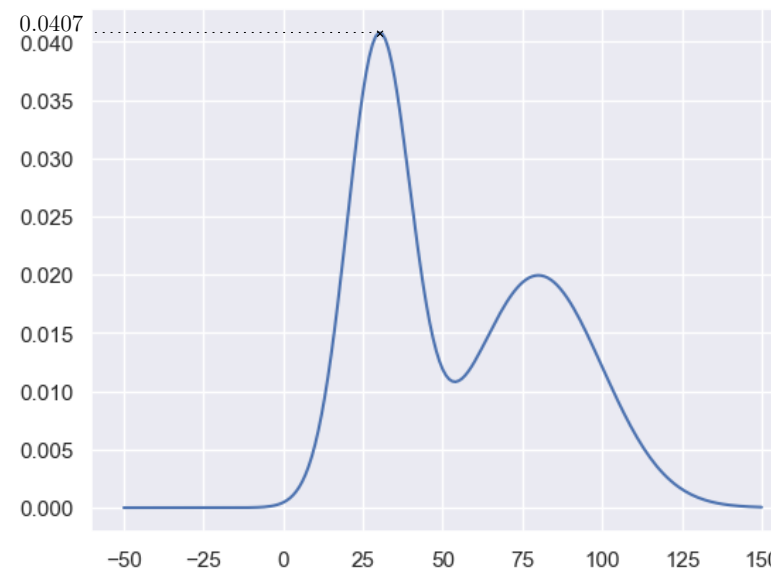
$$p(z_1 \mid z_2, z_3) \propto z_1^{z_2+a-2}(1-z_1)^{b-1} \mathbf{1}_{\{z_1 \geq z_3\}} \sim \text{Beta}(z_1 \mid z_2 + a - 1, b) \mathbf{1}_{\{z_1 \geq z_3\}}$$

$$p(z_2 \mid z_1, z_3) \propto z_1^{z_2} \frac{1}{z_2!} \sim \text{Pois}(z_2 \mid z_1)$$

$$p(z_3 \mid z_1, z_2) \propto \mathbf{1}_{\{z_3 \in [0, z_1]\}} \sim \text{Unif}(z_3 \mid (0, z_1))$$

Some Practice Problems (cont.)

3. Consider the pdf $p(x)$ shown in the following figure.

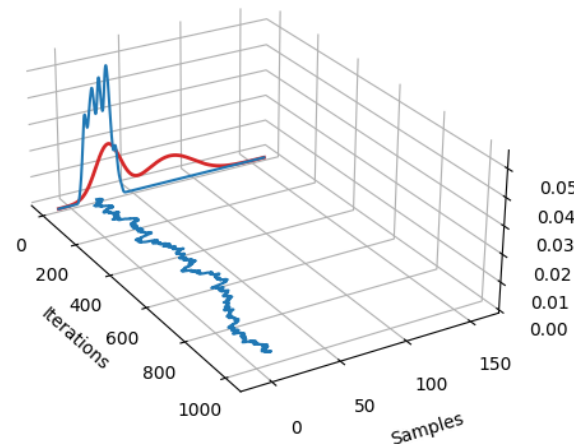


- a) Julia wishes to sample from the truncated distribution whose pdf is $p_1(x) \propto p(x)\mathbf{1}_{\{0 \leq x \leq 125\}}$. Explain how she can use rejection sampling with the uniform distribution as the proposal distribution.
- b) Freddy wishes to find the tail probability $\int_{100}^{\infty} p(x) dx$. Explain how this can be achieved using importance sampling.

Some Practice Problems (cont.)

- c) Russell proposes to use the random walk Metropolis-Hastings algorithm to sample from $p(x)$ using the proposal distribution $q(x, y) = \mathcal{N}(y - x | 0, 1)$. The sampling trajectory is as shown in the following. What is wrong with this approach? What can be done to rectify this?

MH with $N(0, 1^2)$ proposal



Solution:

- a) Let $\tilde{p}_1(x) = p(x)\mathbf{1}_{\{0 \leq x \leq 125\}}$. The proposal distribution can be $q(x) = \text{Unif}(x | 0, 125)$ and $k = 0.0407/125^{-1} = 5.0875$ so that $kq(x) \geq \tilde{p}_1(x)$ for all x .

Some Practice Problems (cont.)

- b) A proposal distribution like $q(x) = \lambda \exp(\lambda(x - 100)) \mathbf{1}_{\{x > 100\}}$ can be chosen. We draw samples $z_1, z_2, \dots, z_n$ from $q(x)$ and compute $w(z_i) = p(z_i)/q(z_i)$. Finally, we compute the estimate as $\frac{1}{n} \sum_{i=1}^n w(z_i)$.
- c) The standard deviation used in the random walk MH is too small. We can increase it so that the acceptance rate is 0.25 to 0.5. E.g., using s.d. of 15, we have the trajectory in the following.

MH with $N(0, 15^2)$ proposal, acc.=0.26

